Glycine is one of the twenty essential amino acids that work in our bodies through many different chemical interactions. It can first be used through its carrier function in nutrient absorption (Hartle et al., 2016). As the smallest and most versatile amino acid, it is an effective ligand for transporting essential minerals into the bloodstream (Lynch, 2004). Glycine can also help regulate brain signaling switches and demonstrates a key role in this (Remko & Rode, 2006). Incorrect binding could lead to neurological errors that could lead to seizures, paralysis, and even death. Lastly, zwitterionic glycine can isolate toxic heavy metals in the blood and help remove them from the body (Remko & Rode, 2006).

Glycine can exist as a zwitterion and a non-zwitterion. A zwitterion is a molecule that contains an equal amount of positively and negatively charged functional groups while still having an overall net charge of zero. On the glycine molecule, the amine group becomes protonated after the carboxyl group donates a proton to it. The positive and negative poles of the zwitterionic form of glycine help create a much stronger electric field within the molecule, which can make a metal ion bind more tightly to it upon contact. In the presence of water, glycine exists as a zwitterion. The intermolecular forces in the presence of water help stabilize the molecule. Knowing this, any model that wants to mimic glycine acting in the body must account for the solvation of water in the computation.

Since the computation needs to use water solvation to accurately represent the interactions going on in the body, there can be two ways to model this effect: the first being implicit solvation and the second being explicit solvation. The implicit solvation model describes how a solvent (in this case, water) affects a molecule without modeling every water molecule in the actual structure. The implicit solvation creates the electric field that water gives off without the hydrogen bonding IMFs. Implicit solvation is primarily used as it requires low computational cost and provides a fast turnaround for calculations. The explicit solvation effect looks at the individual water molecules in the structure and how they interact with the molecule. This model provides a more accurate representation of what is happening inside the body than the implicit model. Its limitations arise when thinking about the higher computational cost and the longer time required to solve the Schrodinger equation. Another issue that arises when using the explicit model is determining where to insert the individual water molecules and how many water molecules to add to the structure.

For my calculations that I did for my research project, I used certain Hamiltonian operators and basis sets to solve the Schrödinger equation. I used B3LYP with D3BJ dispersion for my method, and I used cc-pVTZ for my basis set as my wave function. I then used ORCA and Python to compute and analyze my data. In the research paper, they often used B3LYP or 6-31G* as their basis sets (Remko & Rode, 2006). The calculation used to find the binding energy was similar in the research paper and my research, with it being $\Delta E = C_2H_5NO_2M^{n+} - (M^{n+} + C_2H_5NO_2)$, where $C_2H_5NO_2M^{n+}$ is the metal-glycine complex, M^{n+} is the metal cation, and $C_2H_5NO_2$ is glycine in either the zwitter or non-zwitter form.

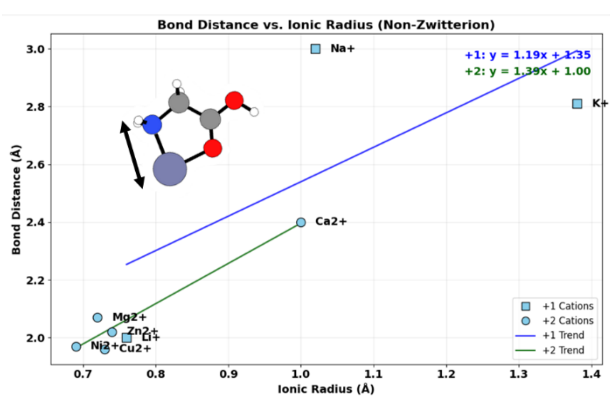

Figure 1. Geometric Correlation in Non-Zwitterionic Complexes.

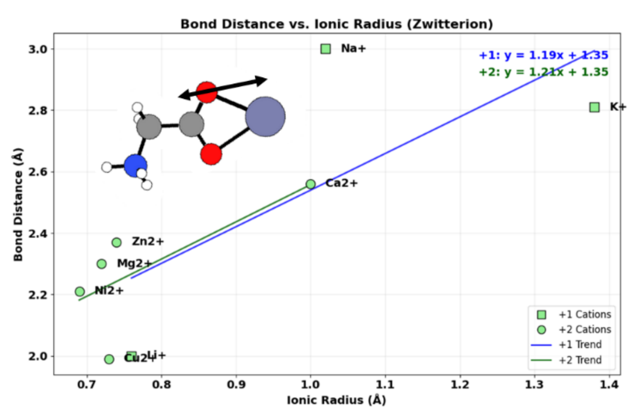

Figure 2. Geometric Correlation in Zwitterionic Complexes.

Table 1: B3LYP/6-311+G(d,p) Optimized Relevant Bond Lengths (Å), Bond Angles (deg), and Dihedral Angles (deg) for the Metal-Coordinated and Hydrated Glycine Species

	$NH_2CH_2COOH{\cdot}M(H_2O)_n$								
parameters	$(H_2O)_n$	Li^+	Na^+	K^+	Mg^{2+}	Ca^{2+}	Ni^{2+}	Cu^{2+}	Zn^{2+}
d[N(6)···M(5)]	$n = 0$	2.062	2.464	2.916	2.123	2.494	1.902	1.988	2.031

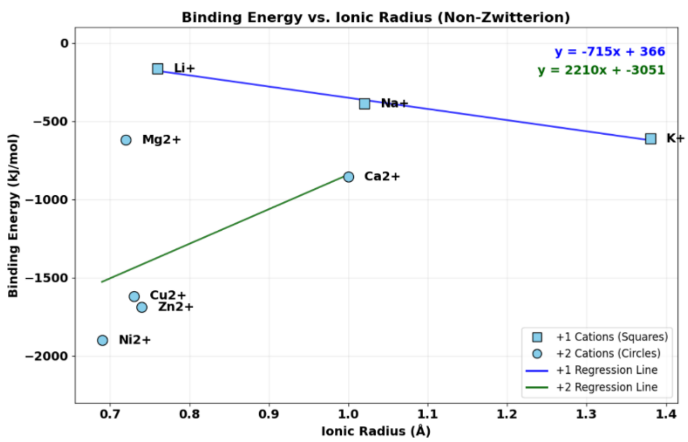

Figure 3. Thermodynamic Affinity of Non-Zwitterionic Glycine Complexes.

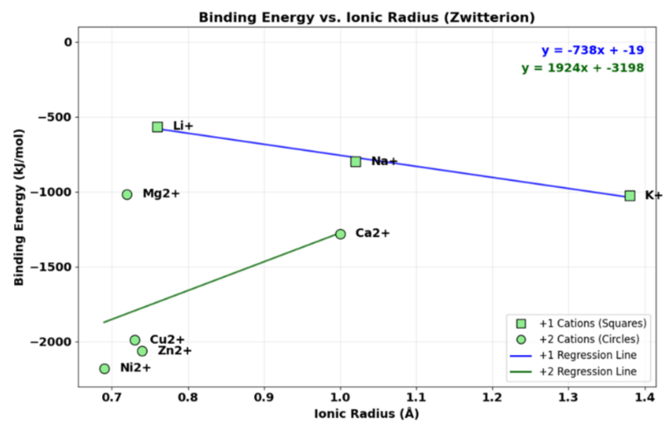

Figure 4. Thermodynamic Affinity of Zwitterionic Glycine Complexes.

Table 3: Calculated Gas-Phase Enthalpies, ΔH, Entropies, ΔS, and Gibbs Energies, ΔG, of the Ion−Glycine Systems

no	system		ΔH^{298} kJ mol^{-1}	ΔS^{298} J mol^{-1} K^{-1}	ΔG^{298} kJ mol^{-1}
1a	Gly···Li^+	A[a]	−250.0	−115.0	−215.7
		B[b]	−257.1	−115.5	−222.7
		C[c]	−236.0	−115.1	−201.7
		exp.[d]	−213.4		−176.1
1b	Gly···Na^+	A	−172.5	−116.1	−137.9
		B	−182.4	−113.4	−148.6
		C	−150.9	−116.3	−116.2
		exp.[d]	−153.1; −157.3; −154.4; −161.0		
1c	Gly···K^+	A	−119.7	−108.2	−87.4
		B	−124.2	−105.5	−92.8
		exp.[d]	−125.5		
1d	Gly···Mg^{2+}	A	−668.5	−130.6	−629.6
		B	−683.9	−125.8	−646.4
		C	−643.9	−130.7	−604.9
1e	Gly···Ca^{2+}	A	−470.0	−115.2	−435.7
		B	−430.7	−113.3	−396.9
1f	Gly···Ni^{2+}	A	−1114.6	−130.3	−1075.8
		B	−1147.3	−153.1	−1101.7
1g	Gly···Cu^{2+}	A	−972.9	−128.4	−934.6
		B	−1001.5	−119.7	−905.8
1h	Gly···Zn^{2+}	A	−846.6	−135.9	−806.1
		B	−872.3	−132.2	−832.9

When comparing the results between my work and the literature, both studies show that 2^+ cations bind stronger than 1^+, and smaller radii bind stronger than larger ones (see Table 1 and Figures 1 & 2). Comparing Table 1 to Figures 1 & 2, Table 1's data shows approximately the same, if not shorter, bond lengths as my data.

Next, we compare binding energies between the explicit and implicit solvation models. The explicit model in the paper showed greater binding affinity than the implicit model that I utilized; however, their choice of the functional they used could also make the binding energies higher than those I obtained (see Figures 3 & 4 and Table 3). Also, they computed the Gibbs Free Energy for each molecule compared to my research, which calculated the internal energy. Since there is no straight comparison of my data with the researchers' data, instead I looked at the research paper's change in enthalpies and compared that to my change in internal energies and looked for any similar trends. Comparing the 1^+ metal-glycine complexes, the research paper shows that from lithium to sodium to potassium, the energy decreased, while in my data, it showed the opposite trend of increasing. For the 2^+ metal-glycine complexes, a similar trend in decreasing energy was shown, starting with nickel and ending with magnesium and calcium being less stable. With the explicit water model molecules being able to hydrogen bond compared to the implicit water model molecules, these extra IMFs likely stabilized the molecule more, making the binding affinity increase.

When comparing the two models to each other, implicit and explicit models recover similar distances between the metal and the glycine regardless of whether it is in the zwitter or non-zwitter form. When comparing the binding energies between the two models, the $^{2+}$ metal complexes show similar trends qualitatively, with the explicit model having more stable binding energies quantitatively. The 1^+ metal complexes, however, show the opposite trend when comparing the two. While implicit solvation models are good for quick trends in what is going to happen, explicit models are definitely necessary for more accurate energy values for 1^+ metals. Since water is not just an electric field and participates in bonds, it must be modeled accurately in the computation to obtain correct data. Therefore, there is a substantial difference between the two models. The binding energies change significantly when including five explicit waters in the structure and show that the bulk water model from the implicit model misses the chemistry needed for accurate representation.

Works Cited

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